Examiner — Complete System Architecture & Workflow

1. Project Overview

Project Name

Examiner

Project Type

AI-Powered Virtual Interview Platform

Development Category

- Full Stack AI Application
- Conversational AI System
- LLM-Powered Platform
- Generative AI Application
- Multimodal AI System

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2. Problem Statement

Existing Problem

Students prepare for interviews using:

- YouTube videos
- Notes
- PDFs
- Static MCQ platforms
- Traditional coding websites

But these methods have several limitations.

Problems in Existing Systems

1. No Real Interview Experience

Most platforms only provide:

- MCQs
- static questions
- theory content

They do not simulate real interviewer interaction.

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2. No Communication Practice

Students may know concepts but fail to:

- explain properly
- communicate confidently
- answer dynamically

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3. No Confidence Analysis

Current systems cannot analyze:

- nervousness
- hesitation
- speaking confidence
- eye contact
- behavior

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4. No Personalized Interview Flow

Most interview preparation systems use:

- fixed questions
- static test patterns

They cannot dynamically adapt based on candidate answers.

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5. Lack of Realistic Environment

Students do not experience:

- pressure of real interviews

- human-like interaction
- conversational flow

Because of this, many students fail real interviews despite having technical knowledge.

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3. Proposed Solution

Examiner

Examiner is an AI-powered conversational interview platform that simulates realistic technical interviews using:

- Generative AI
- NLP
- Speech Processing
- Computer Vision
- Real-time Interaction

Core Idea

The system behaves like a professional interviewer.

It:

- asks interview questions
- evaluates answers
- generates follow-up questions
- analyzes communication
- measures confidence
- generates performance reports

Main Objective

To provide students with:

- realistic interview experience
- technical preparation
- communication improvement
- confidence building
- AI-based feedback

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4. System Workflow

Complete User Flow

```
User Opens Examiner
        ↓
Animated AI Interviewer Appears
        ↓
User Selects Subject
        ↓
Interview Starts
        ↓
AI Asks Question
        ↓
Candidate Answers
        ↓
AI Evaluates Response
        ↓
AI Generates Follow-Up Question
        ↓
Interview Continues
        ↓
Final Report Generated
```

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5. Actual Working of Examiner

Step 1 — Landing Page

When user opens the application:

System displays:

- animated AI interviewer
- subject input field
- interview settings

Example:

```
Subject: Java
Difficulty: Medium
Interview Type: Technical
```

Step 2 — Interview Initialization

User clicks:

```
Start Interview
```

Backend creates:

- interview session
- user context
- AI prompt context

Step 3 — AI Interviewer Activation

Backend sends prompt to Gemini API.

Example Prompt:

```
Act as a professional Java interviewer.
Ask one technical question at a time.
Evaluate answers professionally.
```

Gemini generates first question.

Example:

```
What is JVM?
```

Step 4 — Candidate Response

Candidate answers using:

- text OR
- voice

Step 5 — NLP & AI Processing

The system processes:

Text Analysis

Gemini analyzes:

- correctness
- technical depth
- communication quality

Speech Analysis

Whisper converts speech → text.

Librosa analyzes:

- confidence
- pauses
- voice stability

Computer Vision Analysis

OpenCV + MediaPipe analyze:

- eye contact
- facial movement
- posture
- attention

DeepFace analyzes:

- emotion
- nervousness
- stress

Step 6 — Dynamic Follow-Up Question

Based on answer quality:

If answer is weak

AI asks easier/follow-up questions.

Example:

```
Can you explain bytecode in more detail?
```

If answer is strong

AI increases difficulty.

Example:

```
Explain JVM memory architecture.
```

Step 7 — Final Report Generation

At interview end:

System generates:

- technical score
- communication score
- confidence score
- strengths
- weak areas
- improvement suggestions

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6. Communication Between Components

Overall Communication Architecture

```
Frontend
    ↓
Spring Boot Backend
    ↓
Python ML Service
    ↓
Gemini API + NLP + CV Models
```

Detailed Communication Flow

FRONTEND → BACKEND

Frontend sends:

- subject
- user answer
- voice input
- webcam frames

Backend receives and manages:

- interview state
- sessions
- authentication
- orchestration

BACKEND → GEMINI API

Backend sends:

- interview prompts
- candidate answers
- conversation history

Gemini returns:

- questions
- evaluations
- feedback

BACKEND → ML SERVICE

Backend sends:

- voice/audio
- webcam data
- candidate responses

ML Service performs:

- NLP analysis
- confidence analysis
- emotion detection

ML SERVICE → BACKEND

ML service returns:

- confidence score
- eye contact score
- emotion analysis
- communication metrics

BACKEND → FRONTEND

Backend sends:

- AI questions
- live responses
- interview feedback
- final report

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7. Required Project Folders

Initial MVP Structure

```
Examiner/
│
├── Examiner-frontend/
└── Examiner-backend/
```

This is enough for:

- text interview
- Gemini integration
- AI questioning
- feedback system

Advanced Architecture Structure

When adding:

- NLP
- OpenCV

- Speech AI
- Confidence Analysis

Then use:

```
Examiner/
│
├── Examiner-frontend/
├── Examiner-backend/
└── Examiner-ml/
```

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8. Responsibilities of Each Folder

A. Examiner-frontend

Purpose

User Interface Layer

Technologies

- React
- Tailwind CSS
- Framer Motion
- React Three Fiber

Responsibilities

Frontend handles:

- landing page
- animated interviewer
- user interaction
- voice recording
- webcam access
- interview UI
- final report dashboard

B. Examiner-backend

Purpose

Main System Controller

Technologies

- Spring Boot
- JWT
- MongoDB/PostgreSQL

Responsibilities

Backend handles:

- APIs
- authentication
- interview sessions
- Gemini communication
- report generation
- orchestration
- frontend communication

C. Examiner-ml

Purpose

AI/NLP/Computer Vision Processing Layer

Technologies

- Python
- FastAPI
- OpenCV
- MediaPipe
- Whisper
- DeepFace
- Librosa
- spaCy

Responsibilities

ML service handles:

- speech processing
- emotion detection
- eye tracking
- confidence analysis
- NLP analysis
- behavioral analytics

9. AI Models Used in Examiner

Model/Tool	Purpose
Gemini API	Interview intelligence
Whisper	Speech-to-text
Coqui TTS	AI voice generation
OpenCV	Webcam analysis
MediaPipe	Eye/head tracking
DeepFace	Emotion detection
Librosa	Confidence analysis
spaCy	NLP processing

10. Technology Stack

Frontend

- React
- Tailwind CSS
- Framer Motion
- React Three Fiber

Backend

- Spring Boot
- Spring Security

- JWT
- WebSocket

Database

- MongoDB OR

- PostgreSQL

AI/ML

- Gemini API
- Python FastAPI
- OpenCV
- Whisper
- DeepFace
- MediaPipe

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11. Development Phases

PHASE 1 — MVP

Build:

- landing page
- subject input
- Gemini integration
- AI questions
- text interview
- final report

Folders required:

```
frontend + backend
```

PHASE 2 — Voice AI

Add:

- speech-to-text

- AI voice response
- real-time interaction

Folders required:

```
frontend + backend + ml
```

PHASE 3 — AI Behavioral Analysis

Add:

- OpenCV
- emotion detection
- confidence analysis
- eye tracking

Folders required:

```
frontend + backend + ml
```

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12. Final System Architecture

```
Candidate
    ↓
Frontend (React)
    ↓
Spring Boot Backend
    ↓
Gemini API
    ↓
Python ML Service
    ↓
OpenCV + NLP + Speech Models
```

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13. Innovation of Examiner

Unlike traditional platforms:

Examiner provides:

✅ Real-time AI interview ✅ Dynamic questioning ✅ Conversational interaction ✅ Communication analysis ✅ Confidence analysis ✅ Emotion detection ✅ Personalized feedback ✅ Realistic interviewer experience

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14. Final Conclusion

Examiner is a Full Stack LLM-powered conversational AI platform designed to simulate realistic technical interviews using Generative AI, NLP, Speech Processing, and Computer Vision technologies.

The system combines:

- AI interaction
- voice intelligence
- behavioral analysis
- dynamic interview generation
- full stack architecture

to create an advanced AI-based interview preparation experience.